

1. curly braces make body of method
2. Every statement must ends with semicolon
3. Java is case sensitive `main()` & `Main()` are different

Comments :-

Comments are used to explain or describe

PT

// → single line command

For longer lines `/* */`

`/** */`

↓
Third type of comment

`/* */` → not next in java
if code contains `*/`
code might not deactivate



Java is strongly typed language
Every thing has data type

Hexadecimal 0x

Octal 0

Binary 0b 0x 0B

You can add underscores

1_000_000

✓
humaneyeonly

java compiler removed them

Java does not have unsigned
versions of long, int, short, byte

reduce
complexity



C/C++

long $\rightarrow$ 4 byte 32 bit processor

long $\rightarrow$ 8 byte 64 bit processor

Difficult in making cross platform programs

Java uses fixed size

-X:

Byte to unsignedInt(b)

80
-128 to 127 $\rightarrow$ 0 to 255

chance that integer will not be negative

float $\rightarrow$ 6-7 significant digits $\rightarrow$ 84% x Four

double $\rightarrow$ 15 significant digits

$0.125 = 2^{-3}$ can be expressed

as $0x1.0p^{-3}$

Floating point $\rightarrow$ IEEE 754
Three special floating point values
Overflows & errors

1. NaN
2. positive infinity
3. negative infinity

divergence:-

$$\frac{1}{0} = \infty \quad \frac{0}{0} = \text{NaN}$$
$$\frac{-1}{0} = -\infty$$

Double - POSITIVE INFINITY
NEGATIVE INFINITY
NaN

Corresponds float
(double)

You cannot test

- ~~Dev~~ $x == \text{Double.NaN}$
 $\downarrow$
never true

Double.isNaN(x)
✓

floating point numbers
→ should not be

0.8999

↓

0.9 you would not expect

No precise binary representation of $1/10$

Precision Computations without knowledge → Big Decimal class

Char Type

Data type stores single characters
enclosed in single quotes

'A' → single quote → char

"A" → double quote → string

Originally char → hold characters

Unicode → beyond 65536 characters

some characters → 1 char

Latin letters,
digits

some → 2 chars

Rare emoji

Char → cannot represent every possible

Every character

[\u0000 to \uffff] $\rightarrow$ Unicode.

\u $\rightarrow$ escapes work outside
strings & chars

\n $\rightarrow$ newline

\t $\rightarrow$ tab

\\ $\rightarrow$ Backslash

\" $\rightarrow$ Double Quote

' $\rightarrow$ Single Quote

\b $\rightarrow$ backspace

CAUTION

Unicode processed first before
code processes or parses

\u0022 + \u0022

" + "

\u0022 $\rightarrow$ intended

// \u000A $\rightarrow$ newline

\n $\rightarrow$ newline
X syntax error
breaks comment

//

You
in

2
9

// 1c
↓
found

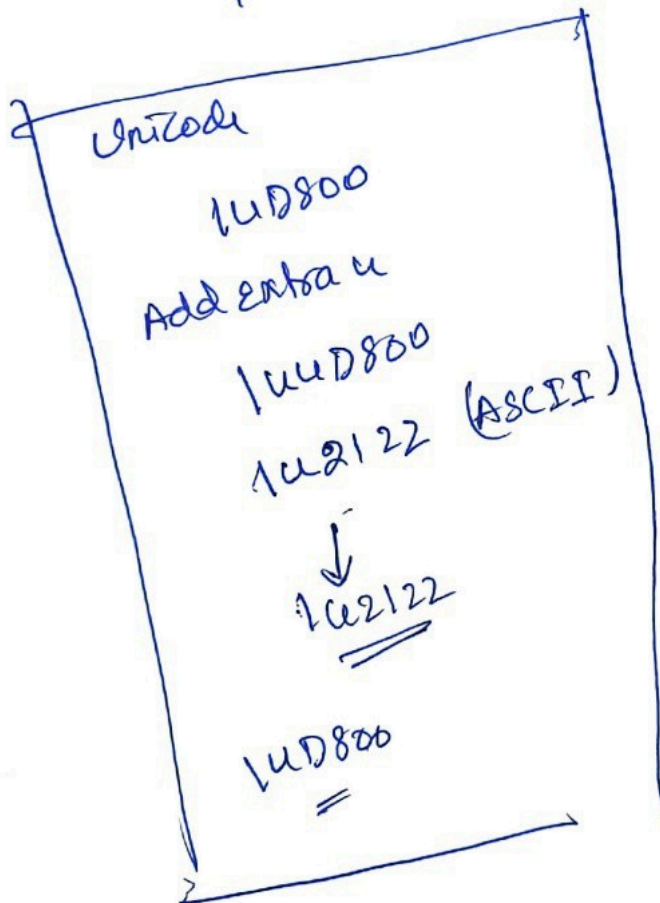
Java
↓
Read hex digits

Syntax Error

You can have any number of 1c
in Unicode

1u4022

1uu4022



So they can
convert
ASCII → Unicode
↑
reverse

Unicode → ASCII
reverse

for ex -
1u00e9 → é

1400e9

